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| Exp No: 2 | Study of Network cables - straight-through an |
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## AIM:

To study the different types of network cables and to make a straight-through cable and a cross-wired (crossover) cable using a crimping tool.

## APPARATUS REQUIRED:

- Cat5e / Cat6 UTP cable (about 1 metre for each cable)
- RJ45 connectors (4 numbers)
- Crimping tool with built-in cutter and stripper
- LAN cable tester (main unit and remote unit)

## ALGORITHM / PROCEDURE:

### Part A - Study of network cables

1. Observe the sample twisted pair, coaxial and fibre optic cables available in the lab.
2. Note the outer jacket marking (for example Cat5e, 24 AWG) and the number of pairs.
3. Write down the use, speed and maximum length of each cable type in the table given below.

### Part B - Making the cables

1. Cut the UTP cable to the required length (about 1 m) using the cutter.
2. Remove about 2.5 cm of the outer jacket from both ends with the stripper. Take care not to cut the inner wires.
3. Untwist the four pairs and straighten the eight wires with fingers.
4. For the **straight-through cable**: arrange the wires in `T568B` order on both ends.
5. For the **crossover cable**: arrange `T568B` order on one end and `T568A` order on the other end.
6. Cut the wire tips even so that about 1.3 cm of the wires is left outside the jacket.
7. Hold the RJ45 connector with the clip facing down, push the wires inside till they touch the end of the connector. The jacket should also go inside a little.
8. Put the connector in the crimping tool and press firmly so that the pins cut into the wires.
9. Repeat steps 2 to 8 for the other end of the cable.
10. Insert both ends into the LAN tester and check the LED sequence.

### Types of network cables

| Cable | Connector | Max. length | Speed | Used for |
|---|---|---|---|---|
| Twisted pair (UTP / STP) | RJ45 | 100 m | 10 Mbps to 10 Gbps | LAN, connecting PCs to switch |
| Coaxial (RG-6, RG-59) | BNC / F-type | 185 m to 500 m | 10 Mbps (old Ethernet) | Cable TV, CCTV, old LANs |
| Fibre optic (single / multi mode) | SC / LC / ST | 2 km (multi) up to 40 km+ (single) | 1 Gbps to 100 Gbps | Backbone, long distance links |

| UTP category | Max. speed | Bandwidth |
|---|---|---|
| Cat5 | 100 Mbps | 100 MHz |

| UTP category | Max. speed | |
|---|---|---|
| Cat5e | 1 Gbps | 100 MHz |
| Cat6 | 1 Gbps (10 Gbps up to 55 m) | 250 MHz |
| Cat6a | 10 Gbps | 500 MHz |

| Pin | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| T568A | W/G | G | W/O | B | W/B | O | W/Br | Br |
| T568B | W/O | O | W/G | B | W/B | G | W/Br | Br |

W = white stripe, G = green, O = orange, B = blue, Br = brown

| Cable | Wiring | Used to connect |
|---|---|---|
| Straight-through | T568B to T568B | PC to Switch, PC to Hub, Router to Switch (unlike devices) |
| Crossover | T568B to T568A | PC to PC, Switch to Switch, Hub to Hub, PC to Router (like devices) |

## DIAGRAM:

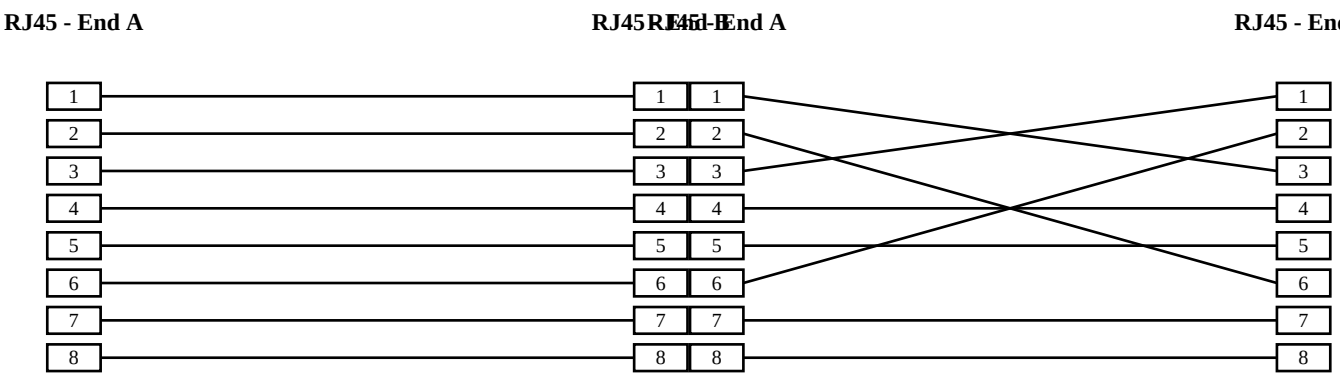


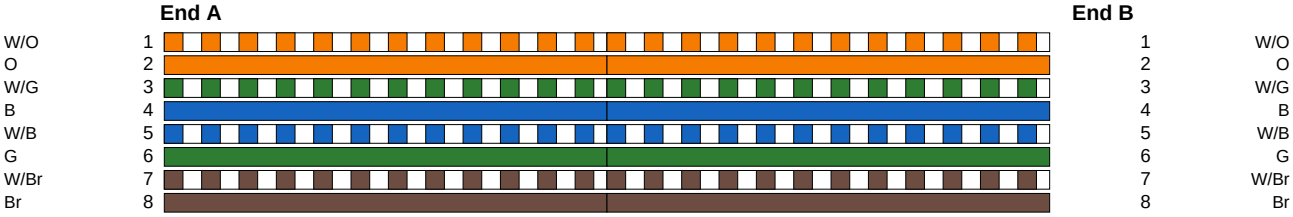

Fig 1: Straight-through cable (pin 1 to 1, 2 to 2 ...)



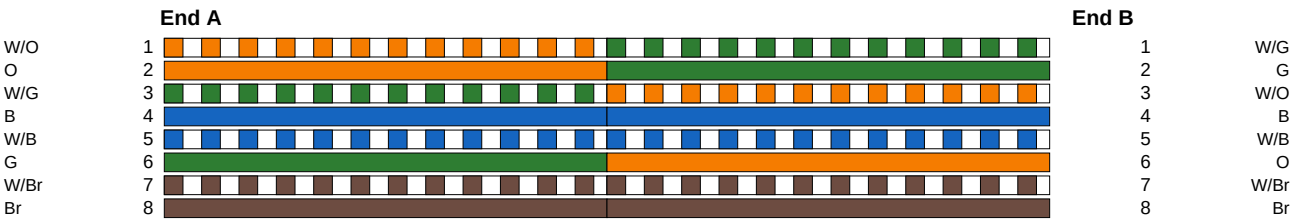

Fig 2: Crossover cable (1 to 3, 2 to 6, 3 to 1, 6 to 2)

## OUTPUT:

### Colour order of the wires at both ends

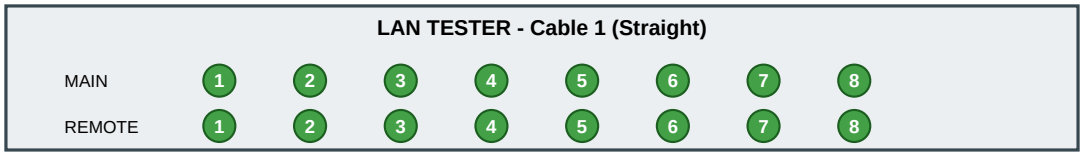



Cable 1: Straight-through cable (T568B - T568B)

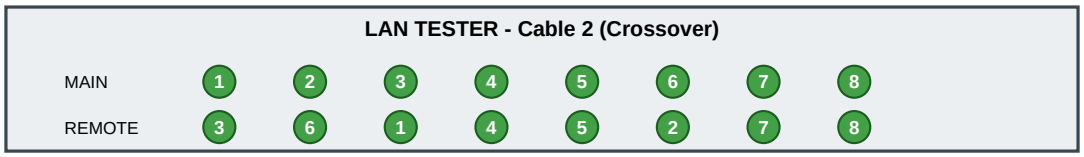



Cable 2: Crossover cable (T568B - T568A)

## LAN cable tester result




**LEDs glow in the order 1-2-3-4-5-6-7-8 : cable is OK**




**LEDs glow in the order 3-6-1-4-5-2-7-8 : crossover cable is OK**

| Cable | Main unit LED | Remote unit LED | Status |
|---|---|---|---|
| Straight-through | 1 2 3 4 5 6 7 8 | 1 2 3 4 5 6 7 8 | Working |
| Crossover | 1 2 3 4 5 6 7 8 | 3 6 1 4 5 2 7 8 | Working |

## RESULT:

Thus the different types of network cables were studied. A straight-through cable and a crossover cable were made using the crimping tool and both were tested successfully with the LAN tester.